Engaging High School Teachers in Artificial Intelligence Concepts and Applications

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Abstract

High school teachers are the primary drivers of education and adoption of new topics in high schools. This work addresses the effectiveness of an artificial intelligence (AI) community-building workshop designed to change high school teachers' perceptions of AI through AI education. The work focuses on contemporary issues related to AI concepts and applications. A group of high school teachers from local education districts was invited to attend a one-day AI hands-on workshop at our university. The workshop included several AI-related topics and hands-on examples and exercises aiming to introduce AI concepts and tools relevant to pre-college education. The participating teachers were expected to become a part of a collaborative network created to design, develop, and implement novel AI learning modules for high school students. Initial and a post-training surveys have been used to measure the impact of this training and to obtain a better understanding of teachers'/students' readiness for additional hands-on AI experiences and further training. The surveys showed that the teachers gained valuable AI knowledge, AI tools, and attitudes that could help them when introducing AI concepts, applications, and current AI ethical issues to their students. The teachers also explored various AI-based teaching tools that they could use to immediately improve learning outcomes for their students.